Report 1: Project Description Report

Hospital Management System (HMS)

Institution	Kigali Independent University (ULK)
Programme	Master of Software Engineering (MSSE)
Assignment	UML Class Diagram Modelling
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1. Project Title and Business Idea

Project title: Hospital Management System (HMS).

Business idea: A hospital needs an information system to manage its patients, doctors and nurses, departments, appointments, prescriptions, laboratory tests, admissions to rooms, and billing. The proposed system replaces paper registers and scattered spreadsheets with one consistent model of hospital data. In this assignment the system is *modelled* with a UML class diagram; building the working application is not required.

2. Problem Statement

Many hospitals, especially small and medium-sized ones, still keep patient information on paper registers or in disconnected files. This causes recurring problems:

- lost, duplicated or incomplete patient files;
- double-booked appointments and long waiting times;
- slow communication of lab results and prescriptions to the staff who need them;
- dispensing and dosage errors caused by illegible or incomplete handwritten prescriptions;
- unclear room and bed availability when patients must be admitted;
- inaccurate or late billing, which reduces trust and hospital revenue.

As a result, quality of care, staff productivity and financial control all suffer.

3. Purpose of the Proposed System

The purpose of the Hospital Management System is to give the hospital a single, structured and consistent record of its people, services and transactions. Its objectives are to:

- keep one reliable record for every patient and make the history easy to retrieve;
- prevent double-booking by linking patients, doctors and appointment times;
- store prescriptions and lab results in a structured, readable form (medicine, dosage, test, result);
- track admissions and room availability;
- produce accurate bills linked to each patient;
- provide a clear UML class model that can later guide database and software design.

4. Project Scope

4.1 Functions covered

Function area	What the system covers
Patient management	Register patients, store personal and contact details, retrieve patient history.
Staff and department management	Maintain doctor and nurse profiles and assign them to departments.
Appointment scheduling	Schedule, reschedule and cancel appointments; record reason, status and diagnosis.
Prescription management	Issue prescriptions made of one or more medicine items (dosage, frequency, duration).
Laboratory tests	Request tests during an appointment; record status and results.
Admission and room management	Admit and discharge patients, assign rooms, track room availability.
Billing and payment	Generate bills for patients and record their payment status.

4.2 Functions excluded

- Staff payroll, recruitment and other human-resources management.
- Storage of medical images (X-ray, MRI, ultrasound).
- Telemedicine and online consultations.
- Pharmacy stock purchasing and supplier management.
- Processing of insurance claims with external insurers.
- Login accounts and user-role classes (for example receptionist or administrator); security is stated only as a non-functional requirement.
- Development of the working application: the project is limited to modelling and documentation.

4.3 Assumptions

- Each doctor and each nurse belongs to exactly one department.
- Each appointment involves exactly one patient and one doctor.
- An appointment results in at most one prescription.
- A room can host several admissions over time (not necessarily at the same moment).
- Billing is recorded per patient.

5. Main Classes and Their Responsibilities

The model contains 13 classes, grouped as people (Person, Patient, Doctor, Nurse), clinical activity (Appointment, Prescription, PrescriptionItem, Medicine, LabTest, Admission) and hospital organisation and finance (Department, Room, Bill).

Class	Responsibility
Person (abstract)	Abstract superclass holding the identity and contact details shared by every person in the hospital.
Patient	Person receiving care. Central class that links appointments, admissions and bills.
Doctor	Medical practitioner who diagnoses patients, issues prescriptions, orders lab tests and supervises admissions.
Nurse	Supports patient care by recording vital signs and assisting doctors within a department.
Department	Organisational unit (for example Cardiology or Paediatrics) that groups doctors and nurses.

Class	Responsibility
Appointment	Scheduled consultation between one patient and one doctor; stores date and time, status, reason and diagnosis.
Prescription	Medical order issued after an appointment, listing the medicines to be dispensed.
PrescriptionItem	One line of a prescription (medicine, dosage, frequency, duration). Resolves the many-to-many link between Prescription and Medicine.
Medicine	Catalogue entry for a drug (name, form, unit price) that can be prescribed.
LabTest	Diagnostic test requested during an appointment; stores the test type, status and result.
Admission	Inpatient stay of a patient, with admit and discharge dates; supervised by a doctor and assigned to a room.
Room	Physical room or ward with a type, a capacity and an availability status.
Bill	Financial document charged to a patient for hospital services; tracks the total amount and payment status.

6. System Requirements

6.1 Functional requirements

ID	Requirement
FR1	The system shall register a patient and keep the patient's personal and contact details.
FR2	The system shall maintain doctor and nurse profiles and assign each to a department.
FR3	The system shall schedule, reschedule and cancel appointments and prevent a doctor from being double-booked.
FR4	The system shall record the reason, status and diagnosis of each appointment.
FR5	The system shall issue prescriptions containing one or more medicines with dosage, frequency and duration.
FR6	The system shall record lab-test requests and their results.
FR7	The system shall admit and discharge patients and assign them to available rooms.
FR8	The system shall generate bills and record payment status.
FR9	The system shall let authorised staff view a patient's history (appointments, prescriptions, tests, admissions).

6.2 Non-functional requirements

Quality	Requirement
Security and privacy	Patient data shall be accessible only to authorised staff (role-based access).
Data integrity	Every appointment, prescription, test, admission and bill shall refer to a valid patient.
Usability	Common tasks such as booking an appointment shall be completed in a few simple steps.
Reliability	The system shall be available during hospital working hours and keep regular backups.
Maintainability	The design shall be modular so that new classes (for example Insurance) can be added later.

6.3 Users of the system

Patients, doctors, nurses and administrative staff. Patients, doctors and nurses are modelled as classes; administrative staff use the system to register patients, schedule appointments and handle billing but are not modelled as a separate class in this version.

7. Conclusion

This report defined the business idea, problem, purpose, scope and requirements of the Hospital Management System. These decisions are the basis of the UML class diagram presented in Report 2.